

Arkadip Som

+91-8509086687 | arkadipsom@gmail.com | [GitHub](#) | [LinkedIn](#) | [Portfolio](#)

EDUCATION

Indian Institute of Engineering Science and Technology, Shibpur Howrah, West Bengal, India
B.Tech in Civil Engineering; CGPA: 7.99/10.00 (Till 4th semester) *Aug 2024 – May 2028*

Begampur High School Hooghly, West Bengal, India
Higher Secondary: 87.8% | Secondary: 97.8% *Jan 2015 – April 2023*

PROJECTS

Texify | *React.js, Node.js, Express.js, PostgreSQL, Google OAuth 2.0, LaTeX* [Live](#) | [GitHub](#)

- Built a **full-stack resume-building flow** with a multi-step form component that maintains nested state across education, experience, and skills sections and a separate **live preview component** that transforms this state into formatted text on the client side for real-time feedback before submission.
- Implemented the backend as an **Express.js** application following a route, controller and model pattern, with a dedicated Resume model persisted in a **PostgreSQL** database hosted on Neon; protected all resume creation and compilation routes with **Express middleware** that verifies **Google OAuth 2.0** sessions restricted to the institute's student email domain **@students.iiests.ac.in**.
- Deployed a separate containerized compilation service built on a **Docker** image with **TeX Live** installed, which receives the structured resume **JSON** over an **authenticated HTTP request**, generates the corresponding LaTeX source, compiles it to PDF, and returns the result as a **binary blob** response consumed directly by the client for download.

CrispURL | *Node.js, Express.js, PostgreSQL, REST APIs* [Live](#) | [GitHub](#)

- Structured the backend into routes, a controller, and a service layer in Express. When a URL comes in through **POST /api/v1/urls/shorten**, the controller validates it and passes it to the service, which checks the urls table for an existing entry by **long_url**, and if none exists, inserts a new record inside a database transaction and encodes the generated ID into a short code using **Base62** encoding.
- Handled redirects through a **GET /:shortCode** route that looks up the short code using an **indexed column in PostgreSQL**, sends back a **302 redirect** to the original URL, and updates the click count asynchronously afterward so the redirect isn't held up by the write.

Climate Resilience & Strategic Capital Allocation Platform | *Python, Scikit-learn* [Live](#) | [GitHub](#)

- Modeled a composite climate resilience score across heatwave risk, flood risk, water stress, population pressure, and infrastructure readiness by normalizing heterogeneous indicators with **min-max standardization** and forecasting temperature and extreme-event trends through 2040 using **OLS linear regression** in Scikit-learn.
- Built a modular decision-support system (separate risk, forecast, and investment engines) with a **custom prioritization matrix** that translates risk scores and forecasts into **actionable adaptation-fund allocation recommendations** for investment pools up to **INR 20,000 crore**, presented through an interactive Streamlit dashboard.

TECHNICAL SKILLS

Languages: Python, C/C++, JavaScript, HTML, CSS

Frontend: React.js, Tailwind CSS

Backend & APIs: Node.js, Express.js, REST APIs

Databases: MySQL, PostgreSQL, MongoDB

Machine Learning: NumPy, Pandas, Matplotlib, Seaborn, Scikit-learn, Neural Network fundamentals

Developer Tools: Git, GitHub, Vercel, Render, Streamlit

CERTIFICATIONS & ACHIEVEMENTS

Open Source Contributor: GirlScript Summer of Code (GSSoC '26) (Apr 2026)

BCG Strategy Consulting & Introduction to Consulting Simulations: Forage (July 2026)

POSITIONS OF RESPONSIBILITY

Publicity Coordinator, INSTRUO 14.0 (Techno-Management Fest, IIST Shibpur) – led outreach at Jadavpur University Salt Lake campus for 30+ events, with focused promotion of Hacksprint, the flagship hackathon (5L prize pool, 55% of the fest's 9L total)

Member, Society of Civil Engineers (SOCCE) – co-organized technical seminars and departmental events